

For my honors option project, I created a tutorial that demonstrates how to build an interactive choropleth map using Plotly Express. The map visualized the Global Hunger Index (GHI) over time. My motivation for selecting this tool and technique stemmed from an interest in this dataset itself. In addition to Data Science, I am also a Psychology major, which has fostered my interest in researching real world issues that directly impact people. I felt that this dataset matched my interests well. After choosing a dataset, I felt a choropleth map would be the best way to display this type of data. I also wanted to highlight how interactive visualization can enhance understanding of complex datasets and choropleth maps are a clear, intuitive way to represent geographic differences. Plotly's interactivity allows viewers to engage with the data and uncover trends at their own pace. This choice aligns well with Segal and Heer's concept of narrative visualization, where data is not merely presented but explored through interactive storytelling.

The tutorial I developed draws heavily on other principles and discussions from the CMSE402 course as well. Throughout the semester, we emphasized that visualization should go beyond aesthetics to convey meaningful insights while remaining transparent and accessible. I applied the concept of data visceralization, drawn from our readings on data feminism, to ensure that my work acknowledged the human lives behind each data point. The GHI data tracks hunger and malnutrition, making it essential to present the information clearly and empathetically. Drawing from Tufte's design philosophy, I attempted to maintain a focus on the substance of the data, while Cairo's emphasis on data scientists' ability to inform public understanding via visualization informed my choices about color, layout, and interactivity. My tutorial ends by discussing elements present in the final visualization that tie to Wickham's Grammar of Graphics framework. Elements like data, aesthetics, etc. were carefully considered and explained within the notebook so that users could understand not just how to build the map, but why each choice matters.

Developing the tutorial presented some challenges for me. The biggest challenge was that I am not a professor, and it was difficult to place myself in a role other than student. I have a new appreciation for my professors after this process. I really struggled to determine the best way to break the process of building the choropleth map down into steps, and if I am being candid, I

think there is a lot of room for improvement in my tutorial. I started by building the code and then trying to add supplemental markdowns that added context to each cell block. This was a difficult process for me and made me glad to usually be on the student end of this workflow. Another challenge I experienced while creating this tutorial was picking a topic. There are a lot of data visualization tools out there and it took me a long time to pick one that I strongly felt like showcasing.

Despite this being the most challenging step and feeling like there is room for improvement, I am still most proud of the overall structure of the tutorial, which builds the final choropleth map step by step. This progressive approach models the iterative process of developing thoughtful data visualizations. By guiding users through each enhancement such as adding color or introducing interactivity, I aimed to demystify what can initially seem like a complex process.

If I had more time, I would expand the tutorial to include features such as interactive filtering or tooltips that provide additional context about each country's GHI score and severity category. I would also have gotten advice from instructors about how they best create their own course tutorials to better decide how to improve my own.